



The situation demands power deficit to be handled at different levels. Managing auxiliary power consumptions plays a vital role in the same" —

N. Venu, president and head, Power Systems Division, ABB Ltd, India



"The task of maintaining quality power is performed very efficiently by power electronics devices such as uninterrupted power supplies (UPS), harmonic filters and other power-quality-enhancing devices. All these devices are a combination of converters/inverters and rectifiers, which are power electronic devices," Prof. Thomas adds.

Transmission and distribution losses: current status

Power transmission losses in India are among the highest in the world at around 4 to 4.5 per cent and a matter of concern for international funding agencies like World Bank. The target for all power plants is to further bring this down by 1 per cent, making it around 3.5 per cent.

As regards distribution loss, ideally it should be confined to a single digit. However, in India this figure hovers around 20-30 per cent (this number is clubbed in the term AT&C losses).

Aggregate technical and commercial (AT&C) losses are the total of technical losses, commercial losses and the shortage due to non-realisation of the

billed amount. All-India AT&C losses are shown in the table here.

"The dream would be to achieve the loss figures of Korea, which stands at the lowest in the world at an enviable 4 per cent. Ministry of Power, government of India, had tied up with the Korean Electric Power Corporation (KEPCO) with the aim of bringing down the losses drastically by 15 per cent in the eleventh plan," says Prof. Thomas.

Under RADRP, work is under progress to collect online data of power and energy flow at distribution transformer level. This will help segregate technical and commercial losses. Once we have a clear picture, appropriate action can be planned to handle technical losses as well as focus on commercial losses by improved metering, billing, collection and other associated areas.

Technical loss reduction measures

Power plants today incorporate a mix of technology systems to help counter transmission and distribution losses.

"Reactive power flow in transmission and distribution lines causes many

All-India AT&C Losses

(in per cent)

Year	AT&C losses (Power Finance Corporation estimate)
2004-05	34.33
2005-06	33.02
2006-07	30.62
2007-08	29.45
2008-09	27.74
2009-10	27.15

Source: Central Electricity Authority (CEA)

problems such as voltage reduction, low power factor and increase in the line losses. Compensation devices for supplying reactive power to the loads at the load end—rather than reactive power flowing from the generating stations—will reduce the line currents and hence the line losses, which is proportional to the square of the current flowing. These devices also improve the power factor and voltage profile. The device may be installed primarily for reactive power supply, power factor correction, and increasing the transmission capability of a line or reducing losses," informs Prof. Thomas.

Power electronic devices termed as FACTS (flexible alternating current transmission system) are a family of thyristor-controlled high-power electronic devices, which are used for a variety of applications in transmission and distribution systems.

"FACTS technology enhances the security, capacity and flexibility of power transmission networks. FACTS solutions help power companies increase transmission capacity over existing AC power lines, providing fast voltage regulation, active power control and load flow control in meshed power systems. These help to minimise bottlenecks in existing transmission systems and improve the availability, reliability, stability and quality of the power supply," explains Venu.

FACTS technology is a comparatively inexpensive and fast way to provide more power and control in existing networks, with minimal environmental impact. ABB has a complete



FACTS SVC overview